Report on Evolutionary Model Exploration for Acoustic Scene Classification

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***Abstract*—Acoustic scene classification (ASC) is a fundamental task in environmental audio analysis. We benchmark deep learning methods for ASC across multiple neural network architectures. We first establish a simple multilayer perceptron baseline to guide subsequent model development, and then evaluate an extended multilayer perceptron, convolutional neural networks, and (multi-stream) ensembles. As a comparison, we also evaluate classical machine learning baselines such as random forests and support vector classifiers. Our experiments quantify the effect of spectral feature choices, input representations, and model complexity on classification accuracy. Overall, convolution-based models provide the strongest gains, achieving an accuracy improvement of 18.7 percentage points over the baseline model.**

I. Introduction

Acoustic Scene Classification (ASC) aims to automatically identify the environment in which an audio recording was captured, such as a street, park, or café. ASC has broad applications in acoustic monitoring, bioacoustics, and context-aware devices [1], where understanding acoustic environments can enhance situational awareness and enable adaptive services.

Because Detection and Classification of Acoustic Scenes and Events (DCASE) challenge evaluations rank systems by performance on a fixed protocol, this setup can incentivise metric-driven optimisation. This motivates controlled architectural comparisons in this work. We (me and my espresso machine) systematically investigate multiple model families for ASC, including multilayer perceptrons (MLPs), convolutional neural networks (ConvNets), ensemble models, and classical machine learning (ML) approaches. By comparing these models, we aim to analyse how different architectural inductive biases and representation learning strategies affect generalisation and alignment with the structural properties of acoustic scene data.

We conduct experiments on the development portion of the *TUT Acoustic Scenes 2016–2018* datasets [2]–[4], comprising recordings from 15 distinct acoustic scenes [5], with a total of 4680 training samples and 540 validation samples of fixed length ($10\,\mathrm{s}$), providing 14.5 hours of audio data. We evaluate performance using classification accuracy, assuming mutually exclusive and balanced classes, with validation accuracy on a held-out subset serving as the primary generalisation metric. In addition to accuracy, we compute confusion matrices and per-class precision/recall/F_1 on the validation set. For brevity we only discuss classes selectively.

In the remainder of this work, we describe the baseline feature pipeline and evaluation procedure. Next, we introduce a baseline MLP and progressively explore more advanced ConvNet architectures, ensemble models, and ML approaches. Finally, we present a comprehensive results analysis, discussing key findings, error patterns, and challenges.

II. Baseline System: Features and Pipeline

A. Feature Representation

The feature representation for the baseline system is derived from log-mel spectrograms that are computed by applying a Short-Time Fourier Transform to the audio samples. In the baseline system, a total of 40 mel-bands are utilised, covering the frequency range from $0\,\mathrm{Hz}$ to $22\,050\,\mathrm{Hz}$, corresponding to the Nyquist frequency for audio sampled at $44.1\,\mathrm{kHz}$.

In the ConvNet architectures, we increase the mel resolution to 128 mel bins to obtain a finer-grained time–frequency representation that better preserves narrow-band spectral structure relevant for distinguishing acoustically similar scenes.

B. Training Setup Shared Across Experiments

All models use a common pipeline with shuffled data loaders, optional mono conversion, min-epoch gated early stopping after 100 epochs, and best checkpointing based on validation accuracy. Additionally, the experiment supports multi-stream preprocessing for stereo files.

All models that use the deep-learning pipeline are trained with cross-entropy loss. For frame-wise models, we compute cross-entropy at the frame level. At evaluation time, we aggregate frame predictions by majority vote:

$$\hat{y} = \mathbf{arg\,max}_c \sum_{t=1}^{T} \mathbb{1}[\hat{y}_t = c],$$

where $t \in \{1..T\}$ indexes the frames. For ConvNet-style models that output clip-level logits, we compute cross-entropy directly on the clip logits, and no majority-vote aggregation is necessary. We fit ML models using each estimator's native optimisation criterion (e. g. impurity-based split optimisation for tree ensembles, maximum-margin optimisation for SVCs).

III. Approach 0—Baseline and Extended MLP

A. Baseline MLP

The baseline MLP processes a feature vector of length 200 through two hidden layers, each comprising 50 units. We apply ReLU activations and dropout with a rate of 0.2 between layers. The output is a linear layer consisting of 15 units corresponding to the 15 target classes. We optimise training

using Adam optimiser for stable performance and a default learning rate of 0.001.

The baseline MLP reached a best validation accuracy of 66.1%. An early training plateau indicates capacity limitations, suggesting performance is constrained by model expressiveness rather than feature representation.

B. Extended MLP

To test whether additional depth improves generalisation without introducing premature overfitting, we increased the model depth to 3 hidden layers with 256 units. This increases representational capacity, enabling more complex non-linear combinations of the input features. Increasing the dropout rate to 0.35 alongside the model depth showed improved training results. We treated validation-accuracy plateaus as a cue to stop increasing capacity and adjust regularisation.

The extended MLP improved to a validation accuracy of 71.5% in the final training run, confirming that increased depth enhanced representation learning. However, diminishing returns from further scaling suggest that fully connected architectures lack structural priors for locality and translation invariance. As a result, spatial correlations in the spectrogram must be learned from data alone, leading to inefficient parameter use and an early generalisation ceiling. In addition to the overall accuracy of 71.5%, the macro-averaged F_1-score reaches 0.7035. Due to the balanced class distribution, macro and weighted averages are similar, ensuring that reported performances are not biased towards dominant classes.

From the classification report (not shown; computed on the validation set), we observed lowest F_1-scores (31.9–57.5%) for the acoustically homogeneous classes `tram`, `beach`, `grocery_store`, `library`, and `café/restaurant`, suggesting that the MLP lacks the inductive bias required to disentangle fine-grained spectral-temporal structures. The `beach` class shows high precision (0.83) but low recall (0.28), reflecting that the model makes few false positives at the expense of missed detections. Figure 1 (confusion matrix) visualises the corresponding confusions among these acoustically similar scenes.

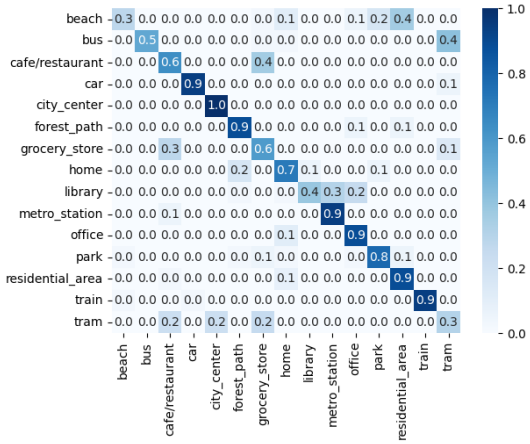

Fig. 1. Class confusion matrix for the extended MLP architecture (rows: true classes, columns: predicted classes).

IV. Approach 1—ConvNet

A. Motivation

While the extended MLP demonstrated that increased depth and capacity can improve performance, a clear plateau in performance was observed.

ConvNets introduce an inductive bias for the previously missing local structure and translation invariance [6]. As log-mel spectrograms exhibit local time–frequency correlations, image-style ConvNet architectures effectively model their spatial hierarchies [7], [8]. This motivates the exploration of a ConvNet-based approach for acoustic scene classification.

B. Architecture

We conduct experiments with a VGGNet-style ConvNet architecture using double convolutional layers and 2×2 pooling [7] to capture wider temporal/frequency context while reducing feature map size. This is inspired by the DCASE 2016 model report by Han and Lee [8]. We apply batch normalisation to each convolutional layer, allowing for the use of a larger batch size of 128, thereby stabilising batch statistics and improving training stability [9]. We use the explicit regularisers Mixup [10] and spectrogram augmentation (SpecAugment) [11] in the following models as task-specific alternatives to conventional regularisation such as dropout [12]. This differs from the reference architectures in [8], [13], which do not incorporate these augmentation-based regularisers.

Mixup generates virtual training examples by linearly interpolating both the input features and their corresponding labels:

$$\tilde{x} = \lambda x_i + (1-\lambda)x_j$$
$$\tilde{y} = \lambda y_i + (1-\lambda)y_j$$
$$\lambda \sim \mathbf{Beta}(\alpha, \alpha),$$

where x_i, x_j are input features from two examples, y_i, y_j are their corresponding labels, λ is a mixing coefficient sampled from a Beta distribution with parameters α, which controls the strength of interpolation. The new data point $(\tilde{x}, \tilde{y})$ is a convex combination of two original examples. SpecAugment applies data augmentation to spectrograms by introducing time masking and frequency masking. Together with VGGNet-inspired dropout in the classifier layer, adding Mixup with an $\alpha \in [0.2, 0.3]$ and SpecAugment with two masks for both time and frequency improved training stability and reduced overfitting in our experiments.

Consistent with Han and Lee [8], who aim at activating units that might otherwise be inactive with ReLU, we use LeakyReLU for activation inputs x_i, defined as

$$y_i = \mathbf{max}(\alpha x_i, x_i),$$

where the slope $\alpha \in (0, 1)$ is set to 0.33. Unlike the classic VGGNet style, we compute activations prior to the convolutions, confirming the observations of Han and Lee [8], and Han et al. [13] regarding steady improvements in accuracy. The

TABLE I
VGGNET-STYLE CONVNET ARCHITECTURE WITH PRE-ACTIVATIONS

Layer	Out Ch.	Kernel / Pool / Notes
Input: $(128 \times T)$ spec	1	Log-mel spectrogram
Block 1	32	
BatchNorm + LeakyReLU		Pre-activation
Conv 3×3		zero-padding=1
BatchNorm + LeakyReLU		Pre-activation
Conv 3×3		zero-padding=1
MaxPool 2×2		Downsampling
Block 2	64	
Block 3	128	
Block 4	256	
Global AvgPool	256	AdaptiveAvgPool2d(1)
Dense (LeakyReLU, Dropout, 512)		Classifier hidden layer
Dense (Softmax, 15)		Scene classes

model's summary can be seen in Table I. We optimise training using stochastic gradient descent with a Nesterov momentum of 0.9, calculating the gradient at a lookahead position. We further increase the learning rate to 0.02, following the guidelines of Han et al. [8], [13]. Although this initially resulted in aggressive and unstable learning, applying a learning rate decay of 0.0001 helped stabilise convergence, as supported by prior DCASE studies, and observed in this work.

C. Interim Conclusion

The ConvNet achieved a best validation accuracy of 79.6%, significantly outperforming the MLP-based models, suggesting that encoding local structure is more effective than increasing fully connected network depth.

Overall, convolutional architectures, combined with VGGNet-style layers, batch normalisation and pooling show the ability to capture richer spectro-temporal patterns, and to generalise better than the previously shown MLPs.

V. APPROACH 2—CONVNET–TCN HYBRID

A. Motivation

The previously evaluated ConvNet models focus on extracting local time–frequency patterns using 2D convolutions. However, acoustic scenes (e. g. `beach` or `city_center`) also contain longer temporal structures spanning multiple frames, such as recurring sound events, background dynamics, and temporal context. Standard 2D ConvNets with pooling progressively reduce temporal resolution, which can limit successful modelling of long-range temporal structure. To address this limitation, we explore a hybrid ConvNet–TCN architecture inspired by [14] and the given VGGNet-style ConvNet setup. We use convolutional layers for local spectro-temporal feature extraction and apply a *Temporal Convolutional Network* (TCN) to model temporal relationships over a wider receptive field with dilated 1D convolutions. This design separates local pattern learning through the ConvNet (in time–frequency space) and temporal context modelling through the TCN (along the time axis).

B. Architecture

The first stage consists of a pre-activated convolutional feature extractor of channel sizes 32 and 64 as shown in Table I. To prepare the representation for temporal modelling, we reduce frequency dimension using adaptive average pooling, resulting in a compact set of frequency-aware feature maps.

TABLE II
TCN TEMPORAL MODELLING BLOCKS

Layer	Out Ch.	Notes
Input: ConvNet features	$C \times F$	Reshape: $(C \cdot F) \times T$
TCN Block 1	64	
Dilated Conv1D		$k = 3$, dilation=1 padding=same
BatchNorm + LeakyReLU		
Dropout		
Dilated Conv1D		$k = 3$, dilation=1
Residual connection		1×1 conv if needed
TCN Block 2	64	
Dilated Conv1D		$k = 3$, dilation=2
BatchNorm + LeakyReLU		
Dropout		
Dilated Conv1D		$k = 3$, dilation=2
Residual connection		
Global AvgPool (time)	64	AdaptiveAvgPool1d(1)
Dense (LeakyReLU, Dropout, 256)		Classifier hidden layer
Dense (Softmax, 15)		Scene classes

After frequency pooling, we reshape the feature maps into a sequence and process them with a TCN. The blocks of the TCN are according to the visualisation in Table II. The use of dilations increases the temporal receptive field without increasing parameters excessively, enabling modelling of longer temporal dependencies. Residual connections stabilise training and facilitate gradient flow. Global average pooling over time produces a fixed-length representation for classification.

C. Architectural Rationale

The ConvNet–TCN model combines complementary inductive biases [14]. The 2D ConvNet learns *local time–frequency structures*. Adaptive pooling compresses frequency dimension while preserving temporal structure. Following that, the TCN models *long-range temporal dependencies*. For that, residual + dilation stabilises training with an expanded temporal receptive field. This expanded receptive field is relevant for scenes that contain recurring background events over several seconds.

D. Interim Conclusion

The ConvNet–TCN hybrid improves validation accuracy marginally by 0.8% to 80.4% (4/540 additional correct clips), so we treat it as an incremental change on this split rather than a decisive improvement. Overall, the result indicates that augmenting a ConvNet feature extractor with explicit temporal modelling may provide small gains on this dataset.

VI. APPROACH 3—MULTI-STREAM CONVNET ENSEMBLE

A. Motivation

The single-stream ConvNet (def. 1-ConvNet) of Section IV operates on one audio representation and may thus overlook complementary spatial and structural cues present in stereo recordings. Environmental sounds often contain spatial, tonal, and transient characteristics that may be suppressed by mono conversion. In contrast to the ConvNet–TCN hybrid, which focuses on temporal modelling, we explore a multi-stream architecture to leverage complementary spatial representations by processing different signal decompositions in parallel.

TABLE III
2-CONVNET CONFIGURATION

Amplitude-normalised audio input	
Channel 1	Channel 2
VGGNet-style ConvNet (Table I)	VGGNet-style ConvNet (Table I)
Concatenate	
Dense (LeakyReLU, 512)	
Dense (Softmax, 15)	

B. Training Setup Adaptations

To limit computational overhead, log-mel spectrogram computation is shared and cached across all streams where applicable. The preprocessing stage decomposes stereo recordings into multiple representations: left/right channels, mid/side encoding, and *Harmonic–Percussive Source Separation* (HPSS) as used by Han et al. [13]. Also following Han et al. [13], we initially train all sub-models separately. We then freeze the sub-models and train a 1024-unit classifier on the concatenated features. In contrast to the referenced paper, however, we do not empirically optimise per-model contribution weights.

C. Stream Definitions

The L and R streams correspond to the original stereo channels. Mid and side streams are defined as $(L + R)/2$ and $(L - R)/2$, representing centre content and stereo differences. Harmonic and percussive streams are obtained using HPSS, capturing tonal and transient components, respectively. Background subtraction algorithms as applied in [13] are not included here, and remain part of future work; nevertheless, background subtraction may support better generalisation by encouraging the model to learn features that are less influenced by spurious factors, such as device-specific background noise.

D. Architecture

The ensemble builds on the VGGNet-style ConvNet by introducing dual-branch ConvNets (def. 2-ConvNet)—see Table III—that processes two log-mel spectrogram inputs while preserving convolutional inductive biases [13]. Branch feature maps are concatenated before the classifier, which operates on the combined representation. We apply audio amplitude normalisation also according to Han et al. [13]. Gathering defined stream definitions and assembling the ConvNets, the ensemble looks like Figure 2.

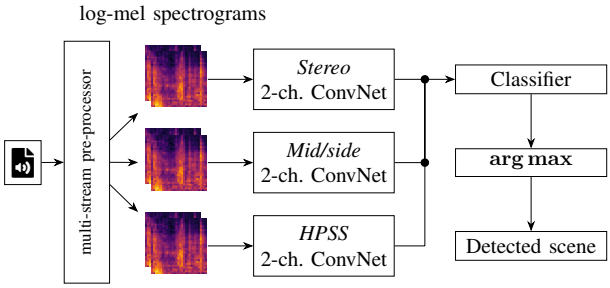

Fig. 2. Ensemble ConvNet model structure. Stacked spectrogram icons indicate two-channel inputs.

E. Interim Conclusion

TABLE IV
ENSEMBLE SUB-MODELS PERFORMANCE OVERVIEW

Model Type	Input Streams	max. ach. val. acc.
Single ConvNet	Mono	79.6%
2-ConvNet	Stereo	79.6%
2-ConvNet	Mid/side	84.8%
2-ConvNet	HPSS	80.9%
ConvNet ensemble	combined	83.3%

We present the ensemble performances in Table IV. The ensemble achieved a validation accuracy of 83.3%, outperforming the 1-ConvNet but slightly trailing the mid/side 2-ConvNet. This result highlights the potential of ensemble learning to improve performance through model diversity. However, it also indicates that input representation quality may exert a stronger influence than model aggregation alone. A systematic exploration of weighted ensemble schemes may help determine whether optimised model contributions can reduce this performance gap.

While ensemble learning improved performance over the 1-ConvNet, the superior results of the mid/side 2-ConvNet indicate that representational advantages can outweigh aggregation-based gains. Beyond its superior validation performance, the mid/side 2-ConvNet branch alone offers a favourable trade-off between accuracy and computational complexity within a single, structurally coherent model. This balance between representational effectiveness and computational efficiency positions the mid/side 2-ConvNet as a standalone solution to deliver both high performance and scalability.

Reaching a validation accuracy of 84.8%, it surpasses the extended MLP's 71.5% by a substantial margin, demonstrating the benefits of leveraging spatial channel decomposition for stereo recordings. Macro- and weighted F_1-scores reach 0.841, indicating consistent performance across most classes. Only `park` ($F_1 = 0.577$) remains a problem class, highlighting the model's strong ability to distinguish between similar acoustic scenes while still struggling with a small subset of recordings.

The confusion matrix in Figure 3 shows substantial improvements in previously problematic categories. In conjunction with the classification report (not shown), classes such as `beach`, `metro_station`, `city_center`, and `forest_path` reach perfect F_1-scores (1.0), whereas the extended MLP had multiple classes below 0.60. This illustrates that the mid/side representation effectively captures spatial and structural cues that fully connected networks fail to model.

Overall, these results confirm that the mid/side 2-ConvNet provides the best trade-off between representational richness, generalisation performance and computational efficiency.

VII. APPROACH 4—CLASSICAL ML BASELINES

The classical ML baseline used features derived from log-mel spectrograms, including basic statistics (e. g. average, minimum, maximum, standard deviation) and higher-order statistics (e. g. skewness, kurtosis, spectral centroid).

A principal component analysis a. k. a. PCA reduced the feature dimensionality to 96 components, followed by a support

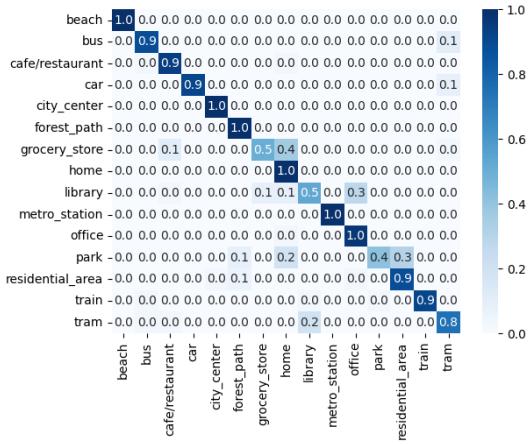

Fig. 3. Class confusion matrix for the mid/side 2-ConvNet architecture (rows: true classes, columns: predicted classes).

vector classifier a. k. a. SVC with a radial basis function kernel for classification, reaching a validation accuracy of 66.7%. Similarly, a random forest model with 2500 estimators and a maximum depth of 28 achieved 65.7%.

To leverage complementary strengths of these models, we use a hard-voting ensemble, combining principal component analysis + support vector classifier and random forest predictions. This ensemble achieved a validation accuracy of 72.6%, outperforming individual ML models and the MLP baseline. Nonetheless, these ML approaches exhibit a representational ceiling when modelling complex spectro-temporal patterns, and further motivate using deep learning strategies for scalable performance improvements.

VIII. Results Overview

The MLP models shown provide computational efficiency but exhibited a clear generalisation ceiling, reflecting the limitations in modelling structured time–frequency patterns.

ConvNet-based models addressed these limitations by exploiting local spectro-temporal structure and achieved substantially higher generalisation performance.

The hybrid ConvNet–TCN architecture, in contrast, emphasised temporal modelling through dilated convolutions and extended receptive fields providing incremental gains, suggesting some benefit from explicit temporal modelling on this dataset.

Classical ML approaches further distinguished themselves with a favourable performance–computation time ratio, and also by demonstrating a slight advantage over MLP-based architectures in terms of validation accuracy when ensembled. Selecting more elaborate feature representations and task-oriented model compositions could yield competitive performance, as demonstrated by Abrol et al. [15] using a Gaussian mixture model and archetypal analysis.

Gathering the results in Figure 4 and including mean validation performances of [13] (HLP17), we conclude this work by choosing the mid/side 2-ConvNet as the contender for the final submission.

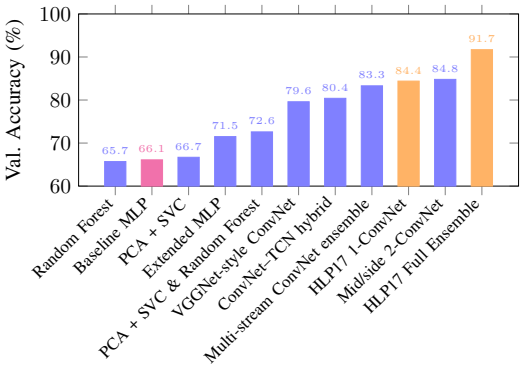

Fig. 4. Bars show the best validation accuracy observed during our experiments; HLP17 values are reported means from the reference. [13].

While the mid/side 2-ConvNet demonstrated the strongest overall performance among the models evaluated and represents the primary candidate for submission of this work, the ConvNet ensemble remains particularly noteworthy. The ensemble revealed the potential benefits of separating signal representations into multiple complementary streams, offering greater controllability of individual feature pathways.

Across all experiments, performance improvements followed a consistent pattern: models that introduced stronger task-aligned inductive biases yielded the most substantial gains. MLPs, lacking spatial priors, reached an early ceiling. Convolutional architectures, encoding locality, and translation invariance, achieved the largest performance jump. Temporal modelling via TCN provided only incremental improvements in our setup, suggesting that long-range temporal context is not a primary bottleneck for these models on this split. Lastly, multi-stream approaches demonstrated that enhancing input representation can outweigh aggregation complexity. These findings suggest that representational alignment played a larger role than architectural scaling in our experiments.

IX. Error Analysis and Discussion

A. Observed empirical limitations

We observed a performance saturation around 70% during early stages in the course of the work, indicating that certain discriminative characteristics of the acoustic scenes were not fully captured by chosen feature representations. Notably, this saturation was particularly evident during early randomised convolutional experiments, which lacked exploitation of relevant audio features. This further supports our conclusion that task-aligned inductive biases and representational alignment are critical for achieving substantial performance gains.

Mixup offers regularisation benefits, but may introduce unrealistic samples in ASC tasks, potentially limiting performance.

B. Practical / implementation insights

Prior DCASE submissions served as methodological references, guiding not only architectural choices but also feature engineering decisions, emphasising that effective ASC

performance depends on a careful alignment between signal representation, preprocessing strategy, and model design.

A critical challenge involved maintaining consistency between acoustic feature extraction hyper-parameters and corresponding model input specifications, where configuration mismatches resulted in significant performance degradation.

C. Future work / methodological expansion

This work demonstrates that ASC performance is not solely determined by model complexity and composition, but by the fine alignment between feature representation, preprocessing strategy, and architectural inductive bias.

With regard to future enhancements, a combination of the 2-ConvNet approach and TCN extension could be used to incorporate the aforementioned advantages into a wider range of extracted information included in the decision-making process. More complex approaches such as the integration of transformers are not explored, but may be feasible on larger datasets. Future work could also reassess transfer learning using large-scale pre-trained audio embeddings (e. g. OpenL3), as used in the DCASE 2021 system of Diez and Saratxaga [16].

We did not perform a fixed number of repeated runs across multiple random seeds for each configuration; therefore, reported validation accuracies should be interpreted as point estimates. Future work should define a fixed n of runs combined with using k-fold cross-validation to provide rigorous statistical performance estimates. This would enhance comparability and reduce sensitivity to single data splits, or deliver information about the optimal model complexity.

Reproducibility

We include all source code required for conducting and analysing experiments in the repository, along with up to date configurations. The repository includes the data preprocessors used and a README with step-by-step instructions.

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